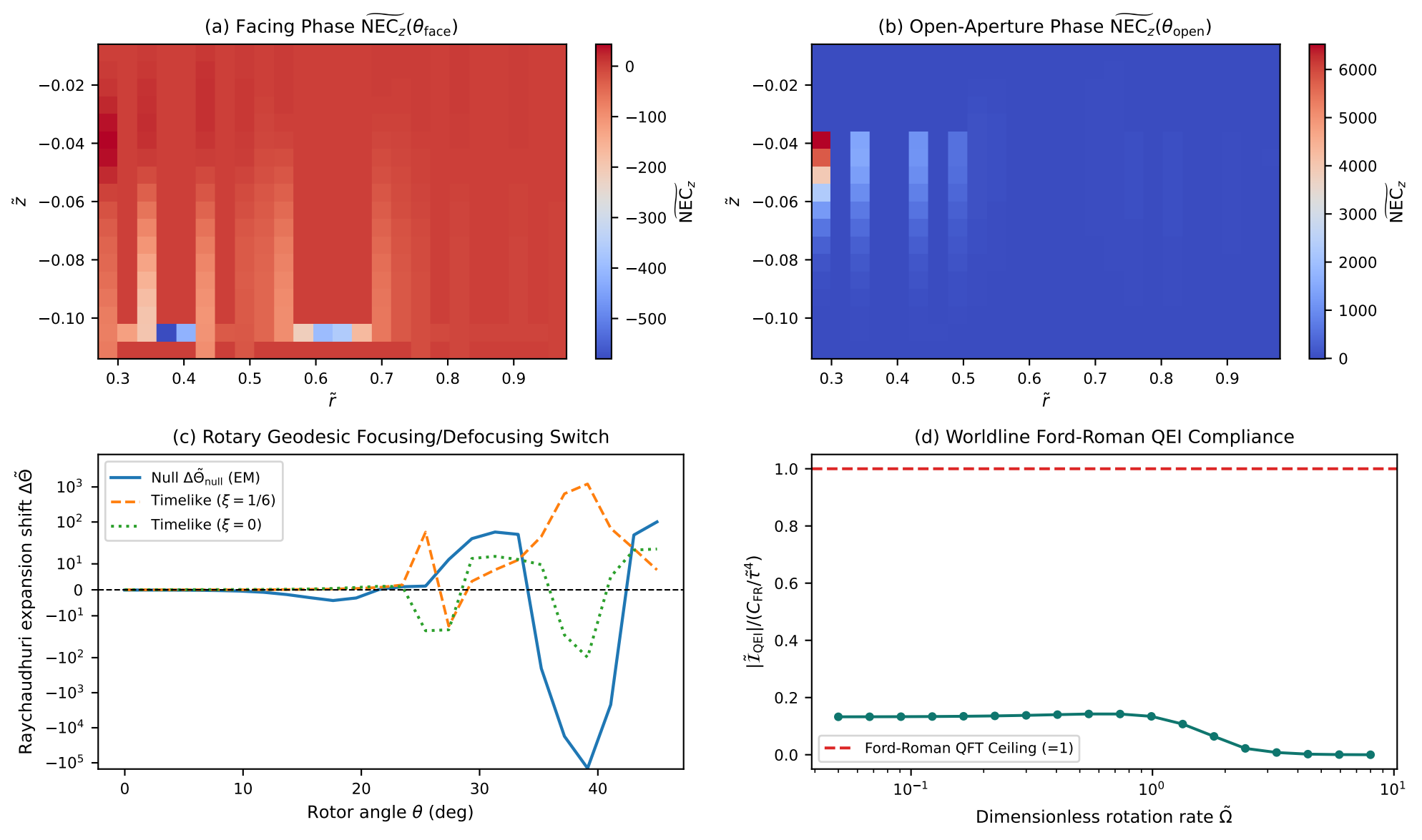
(a) Facing Phase $\widetilde{\mathrm{NEC}_z}(\theta_{\mathrm{face}})$
$\tilde{z}$
$\tilde{r}$
$\widetilde{\mathrm{NEC}_z}$
(b) Open-Aperture Phase $\widetilde{\mathrm{NEC}_z}(\theta_{\mathrm{open}})$
$\tilde{z}$
$\tilde{r}$
$\widetilde{\mathrm{NEC}_z}$
(c) Rotary Geodesic Focusing/Defocusing Switch
Null $\Delta\tilde{\Theta}_{\mathrm{null}}$ (EM)
Timelike ($\xi = 1/6$)
Timelike ($\xi = 0$)
Raychaudhuri expansion shift $\Delta\tilde{\Theta}$
Rotor angle θ (deg)
(d) Worldline Ford-Roman QEI Compliance
$|\tilde{\mathcal{I}}_{\mathrm{QEI}}|/(C_{\mathrm{FR}}/\tilde{\tau}^4)$
Ford-Roman QFT Ceiling (=1)
Dimensionless rotation rate $\tilde{\Omega}$